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PHAR0002: An introduction to the mechanisms of drug action.

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DRUG ANTAGONISM.

This handout has been modified from the original text developed by Prof. D.H. Jenkinson who taught at UCL for many years and eventually converted his notes into an excellent chapter in the *Textbook of Receptor Pharmacology* (Ed., Foreman, J.C., and Johansen, T., CRC Press, 1996/2003/2010).

Aims:

1) To understand the characteristics of reversible competitive antagonism.

2) To be able to understand the principles involved in the derivation of the Schild equation.

3) To remember the Schild equation, the Schild plot and how these can be used to test for reversible competitive antagonism.

Antagonists.

Many of the most useful drugs are antagonists, *i.e.*, substances that reduce the action of another agent which is often an endogenous agonist (e.g., a hormone or neurotransmitter). Though the most common mechanism is simple competition, antagonism can occur in a variety of ways.

Reversible competitive antagonism.

In this type of antagonism either the agonist or the antagonist molecule can interact with the receptor to form a bound drug-receptor complex, but it is not possible for both types of molecule to bind simultaneously. The bonds that hold these receptor-drug complexes together are comparatively weak so the pairing of drug and receptor is transient (thermal energy will, sooner or later, break the pair apart again). Binding between receptors and drug molecules occurs in a random fashion. However, the greater the concentration of drug (agonist or antagonist) the greater the chance of a collision occurring between the drug and a vacant receptor and therefore the greater the chance that a complex will form. For this reason, the balance between agonist-receptor and antagonist-receptor occupancy can be shifted by changing their relative concentrations; adding antagonist will reduce the receptor occupancy by agonist while an increase in agonist will restore the balance. This idea is illustrated in Figure 1.

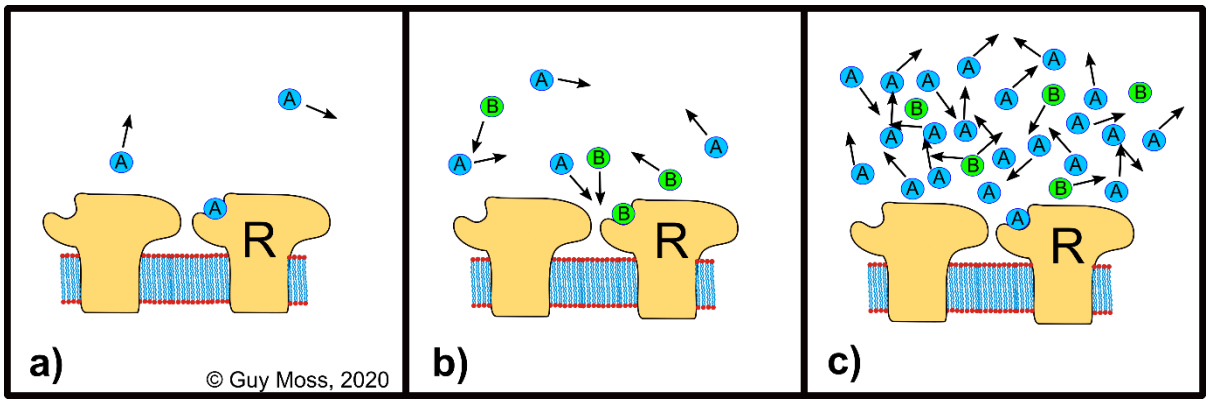


Figure 1. The fraction of receptors (R) occupied by agonist (A) or antagonist (B) depends on the relative concentration of each. (a) When only agonist and receptor are present collisions between receptor and agonist can frequently create a complex – there is an absence of competition for the vacant receptors. (b) When antagonist is added, there is suddenly a competition for binding when a receptor becomes vacant; either agonist or antagonist can occupy it next. Further, otherwise successful attempts by the agonist to bind to a receptor can now fail if the receptor is already occupied by antagonist. (c) Increasing agonist concentration increases the receptor occupancy by agonist molecules because, when receptors become available, they will, on average, be more likely to 'meet' and bind to an agonist rather than an antagonist molecule.

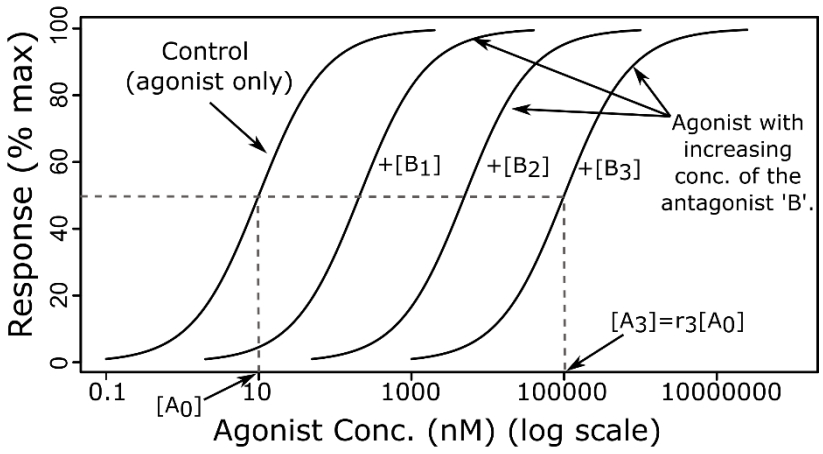


Figure 2. The predicted effect of three concentrations of a reversible competitive antagonist, B, on the response to an agonist. The measurements needed to calculate the concentration ratio r_3 (for the highest concentration of antagonist) is also illustrated.

Thus, as might be expected from our discussion, a reversible competitive antagonist (for example, atropine) alters the concentration-response relationship for the action of an agonist (for example, acetylcholine). We start by examining how a reversible competitive antagonist (for example, atropine) alters the concentration-response relationship for the action of an agonist (for example, acetylcholine). It is found experimentally that the presence of the antagonist causes the agonist log concentration-response curve to be shifted to the right,

without any change in slope or maximal response. The antagonism is *surmountable* over a wide range of concentrations. This is illustrated in Figure 2.

The extent of the shift is best expressed as a ***concentration ratio***[1]. This is defined as the factor by which the agonist concentration must be increased to restore a given response in the presence of the antagonist. The calculation of the concentration ratio is done as follows. First, a certain magnitude of response is selected. This is often 50% of the maximum attainable, but in principle any value would do[2]. In the absence of antagonist, this response is elicited by a concentration $[A_0]$ of agonist. When the antagonist is present, to achieve the same level of response, the agonist concentration has to be increased to a new value say, [A']. This concentration [A'] is greater than $[A_0]$ and so it is possible to find a number *r*, greater than one, such that $[A']=r[A_0]$. In other words, we can view the required increase in agonist concentration as an increase by a factor *r,* i.e., to $r[A_0]$. Thus, for antagonist concentration $[B]_3$ in Figure 2, the ratio of agonist concentration in the presence and absence of antagonist is r_3 (i.e. $[A_3]/[A_0]=r_3$).

The negative logarithm of the concentration of antagonist (in molar units) which causes a concentration ratio of *x* is sometimes denoted by pA_x. This term was introduced by H.O. Schild as an empirical measure of the activity of an antagonist. The value most often quoted is pA_2, where:

$$\mathrm{pA}_2 = -\log[B]_{r=2}$$

To illustrate this notation, we consider the ability of atropine to block the muscarinic receptors for acetylcholine. A concentration of only 1 nM makes it necessary to double the acetylcholine concentration needed to elicit a given submaximal response of a tissue. Hence $pA_2 = 9$ for this action of atropine ($-\log(10^{-9}) = 9$). Notice that by convention we use molar units for this, just in the same way that we do for pH ($= -\log[H^+]$).

We next look at why there should be a parallel shift in the curves, and at the same time we will derive a simple but most important relationship between the amount of the shift, as expressed by the concentration ratio, and the concentration of the antagonist. We assume for simplicity that when the tissue is exposed to the agonist and the antagonist at the same time, the two drugs come into equilibrium with the binding sites on the receptor macromolecules. At a given moment an individual site may be occupied by either an agonist or an antagonist molecule, or it may be vacant. The relative proportions of the total population of binding sites occupied by agonist and antagonist are governed by the concentrations of agonist and antagonist, and by the affinities of the sites for each. Because the agonist and the antagonist combine reversibly with the receptor, raising the agonist concentration will increase the proportion of sites occupied by the agonist, at the expense of antagonist occupancy. Hence the response will increase.

[1] or *dose ratio* - both terms are used.

[2] though clearly it is sensible to avoid the extreme ends of the range.

We begin by assuming that both the agonist, A, and the antagonist, B, combine with their binding site according to the law of mass action, and in a way that can be represented by the two reactions below. We know it is an oversimplification for the agonist, but for the sake of simplicity consider:

$$A + R \rightleftharpoons AR$$

$$B + R \rightleftharpoons BR$$

Our first task is to work out how the proportion of receptors occupied by the agonist varies with the concentrations of the agonist and the antagonist. Equilibrium is assumed. Applying the law of mass action gives:

$$[A][R] = K_A[AR] \qquad \text{eq. (1)}$$

And similarly:

$$[B][R] = K_B[BR] \qquad \text{eq.(2)}$$

where K_A and K_B are the dissociation equilibrium constants for the combination of A and B respectively for the binding sites.

As in earlier lectures, these equations can be rewritten in terms of the proportions of binding sites that are free (p_R) or occupied by either A (p_{AR}) or B (p_{BR}):

$$[A]p_R = K_A p_{AR} \qquad \text{eq.(3)}$$

$$[B]p_R = K_B p_{BR} \qquad \text{eq.(4)}$$

An individual receptor is either vacant or occupied by an agonist or an antagonist molecule. Hence, we can write:

$$p_R + p_{AR} + p_{BR} = 1 \qquad \text{eq.(5)}$$

What we need to know is p_{AR}, so we use equations 3 and 4 to substitute for p_R and p_{BR} in equation 5:

$$\frac{K_A}{[A]} . p_{AR} + p_{AR} + \frac{[B]}{K_B}\frac{K_A}{[A]} . p_{AR} = 1 \qquad \text{eq.(6)}$$

$$p_{AR} = \frac{[A]}{K_A\left(1+\frac{[B]}{K_B}\right)+[A]} \qquad \text{eq.(7)}$$

Termed the ***Gaddum equation***, after J. H. Gaddum, Equation 7 expresses how p_{AR} depends on the concentrations of A and B. If [B] is set to zero, we have the Hill-Langmuir equation.

In order to apply this expression to the practical problem of how a competitive antagonist will affect the response to the agonist, we need to make some assumption about the relationship

that exists between the response and the proportion of receptors occupied by agonist. Gaddum and Schild recognised that the best way to proceed was to assume that the same response (say 30% of the maximum attainable) corresponded to the same receptor occupancy by agonist whether the agonist was acting alone, or in the presence of the competitive antagonist. This assumption makes it unnecessary to know the exact form of the relationship between occupancy and response. This was a most important advance, however obvious it might seem in retrospect.

We can now consider an actual experiment in which a certain response, say 30% of the maximum, is elicited first by a concentration [A] of agonist acting alone, and then by a concentration *r*[A] when the agonist is applied in the presence of the antagonist. Here *r* is the concentration ratio, as already defined.

Since p_{AR} is assumed to be the same in the two situations, we can then write:

$$\frac{[\mathrm{A}]}{K_\mathrm{A}+[\mathrm{A}]} = \frac{r[\mathrm{A}]}{K_\mathrm{A}\left(1+\frac{[\mathrm{B}]}{K_\mathrm{B}}\right)+r[\mathrm{A}]} \qquad \text{eq.(8)}$$

So

$$\frac{[\mathrm{A}]}{K_\mathrm{A}+[\mathrm{A}]} = \frac{[\mathrm{A}]}{K_\mathrm{A}\left(\frac{1+\frac{[\mathrm{B}]}{K_\mathrm{B}}}{r}\right)+[\mathrm{A}]} \qquad \text{eq.(9)}$$

If the expressions on the right and the left are to take the same value, the following equality must hold:

$$\frac{1+\frac{[\mathrm{B}]}{K_\mathrm{B}}}{r} = 1 \qquad \text{eq.(10)}$$

Hence

$$r - 1 = \frac{[\mathrm{B}]}{K_\mathrm{B}} \qquad \text{eq.(11)}$$

This is the ***Schild equation*** which was first applied to the study of competitive antagonism by H. O. Schild in 1949. It is probably the most important single quantitative relationship in pharmacology and has been shown to apply to the action of many competitive antagonists over a wide range of concentrations. Though originally derived (as here) on the basis of the simple scheme for receptor activation described in Lecture 2, it holds equally for the more realistic models considered in later lectures.

One of the predictions of the Schild equation is that a reversible competitive antagonist should cause a parallel shift in the log agonist concentration response curve (as illustrated in Figure 2). This is because for a given value of [B] and of K_B, the concentration ratio, *r*, should also be constant, regardless of the concentration and even the identity of the agonist (provided that it acts through the same receptors as the antagonist). With a logarithmic scale,

a constant value of r corresponds to a constant separation of the concentration-response curves, i.e., parallelism (because log (*r*[A]) - log [A] = log *r* + log [A] - log [A], = log *r,* regardless of the value of [A]).

Perhaps the most important application of the Schild equation is that it provides a way of estimating the dissociation equilibrium constant for the combination of an antagonist with its binding sites. A series of agonist concentration-response curves is established, first without and then with increasing concentrations of antagonist, and tested for parallelism. If this condition is met, the value of (*r* - 1) is plotted against the antagonist concentration. This should give a straight line of slope equal to the reciprocal of K_B.

More usually, both (*r* - 1) and [B] are plotted on logarithmic scales (the ***Schild plot***). The outcome should be a straight line with a slope of unity, and the intercept on the *x* - axis provides an estimate of log K_B. The basis for these statements can be seen by expressing the Schild equation in logarithmic form:

$$\log(r-1) = \log[\mathrm{B}] - \log K_\mathrm{B} \qquad \text{eq.(12)}$$

A Schild plot (based on the results of a student class practical on the effect of atropine on the contractile response of guinea-pig ileum to acetylcholine) is shown in Figure 3. Note that the line is straight, and its slope close to unity, as Equation 12 predicts.

How might the value of pA_2 (see page 3) be interpreted in these terms? *If the Schild equation is obeyed,* pA_2 then gives an estimate of -log K_B, because (from Equation 12)

$$\log(2-1) = \log(1) = 0 = \log[\mathrm{B}]_{r=2} - \log K_\mathrm{B} \qquad \text{eq.(13)}$$

$$\therefore -\log[\mathrm{B}]_{r=2}\,, = \mathrm{pA}_2, = -\log K_\mathrm{B} \qquad \text{eq.(14)}$$

The term pK_B is often used to denote -log K_B.

To summarise, reversible competitive antagonism has the following characteristics:

1. The action of the antagonist can be overcome by a sufficient increase in the concentration of agonist (i.e., the antagonism is surmountable)

2. In the presence of the antagonist, the curve relating the log of the agonist concentration to the size of the response is shifted to the right in a parallel fashion.

3. The quantitative relationship between the magnitude of the shift (as expressed by the concentration ratio) and the antagonist concentration obeys the Schild equation.

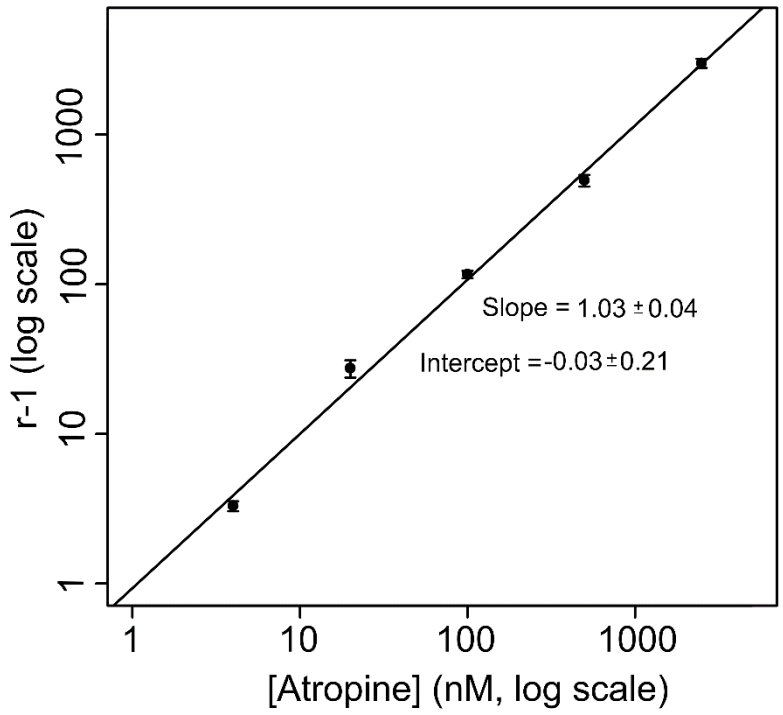


Figure 3. Schild plot for the action of atropine in antagonising the action of acetylcholine on guinea-pig ileum. Each point gives the mean ± the standard error of the mean of the number of observations shown. From a student class experiment at UCL.

Some other mechanisms of drug antagonism.

Many of the most useful drugs are antagonists, i.e., substances that reduce the action of another agent which is often an endogenous agonist (e.g. a hormone or neurotransmitter). Though the most common mechanism is simple competition, antagonism can occur in a variety of ways.

Mechanisms not involving the receptor macromolecule through which the agonist acts.

1. Chemical antagonism. The antagonist combines directly with the substance being antagonised. Receptors are not involved.

> Example: the chelating agent EDTA is used to treat inorganic lead poisoning (a less toxic chelate is formed and excreted).

2. Functional or physiological antagonism The 'antagonist' is actually an agonist which produces an opposite biological effect to the substance being antagonised. Each substance acts through its own receptors. See also indirect antagonism (below)

> Example: adrenaline relaxes bronchial smooth muscle and can thereby reduce the bronchoconstriction caused by histamine and the leukotrienes.

3. Pharmacokinetic antagonism Here the 'antagonist' effectively reduces the concentration of the active drug at its site of action.

> Example: repeated administration of the anti-epileptic drug phenobarbitone induces an increase in the activity of hepatic enzymes that inactivate the anticoagulant drug warfarin. Hence if phenobarbitone and warfarin are given together, the plasma concentration of warfarin is reduced, so that it becomes less active, at a given dose.

4. Indirect antagonism The antagonist acts at a second 'downstream' receptor which links the action of the agonist to the final response observed.

> Example 1. β-adrenoceptor blockers such as propranolol reduce the increase in heart rate caused by indirectly-acting sympathomimetic amines such as tyramine. This is because tyramine acts by releasing noradrenaline from noradrenergic nerve endings, and the released noradrenaline acts on β-adrenoceptors to increase heart rate:

$$\text{tyramine} \rightarrow \text{release of noradrenaline} \rightarrow \beta\ \text{receptor activation} \rightarrow \text{response}$$

Another possibility is that the antagonist interferes with other post-receptor events which contribute to the tissue response.

> Example 2. Calcium channel blockers ('calcium antagonists') such as verapamil block the influx of calcium needed for smooth muscle contraction, and hence they reduce the contractile response to acetylcholine.

Some pharmacologists prefer to describe this as a variant of functional antagonism (see above).

Appendix – Step-by-step derivation of the Gaddum equation.

Starting again from the simple equilibria:

$$A + R \rightleftharpoons AR$$

$$B + R \rightleftharpoons BR$$

Our aim is to find the Gaddum equation which relates the proportion of receptors occupied by agonist, p_{AR}, to the agonist concentration [A], the agonist dissociation equilibrium constant K_A, the antagonist concentration [B] and the antagonist dissociation equilibrium constant K_B.

We start by applying the law of mass action to the above scheme, giving:

$$[A][R] = K_A[AR] \qquad \text{eq. A1}$$

And similarly:

$$[B][R] = K_B[BR] \qquad \text{eq. A2}$$

where K_A and K_B are the dissociation equilibrium constants for the combination of A and B respectively for the binding sites. (If you are uncertain about this first step look back at your notes on the Hill-Langmuir equation.)

Now we define the total concentration of receptor $[R]_T$, which is the sum of receptors in all possible states. The three possible states in the scheme we are considering are R (the free receptor), AR (the receptor with agonist bound) and BR (the receptor with antagonist bound), so:

$$[R] + [AR] + [BR] = [R]_T \qquad \text{eq.A3}$$

If we divide both equations A1 and A2 by $[R]_T$ we get:

$$[A]\frac{[R]}{[R]_T} = K_A\frac{[AR]}{[R]_T} \qquad \text{eq. A4}$$

$$[B]\frac{[R]}{[R]_T} = K_B\frac{[BR]}{[R]_T} \qquad \text{eq. A5}$$

The terms p_{AR} and p_{BR} represent the proportion of receptors occupied by A (p_{AR}) or B (p_{BR}).

They are:

$$p_{AR} = \frac{[AR]}{[R]_T} \qquad \text{eq.A6}$$

and

$$p_{BR} = \frac{[BR]}{[R]_T} \qquad \text{eq.A7}$$

Substituting A6 and A7 into A4 and A5 respectively we get:

$$[\mathrm{A}]p_{\mathrm{R}} = K_{\mathrm{A}}p_{\mathrm{AR}} \qquad \text{eq.A8}$$

$$[\mathrm{B}]p_{\mathrm{R}} = K_{\mathrm{B}}p_{\mathrm{BR}} \qquad \text{eq.A9}$$

Equations A8 and A9 can be rearranged into forms that will be more useful shortly. To do this we divide both sides of A8 by K_A and both sides of A9 by K_B. This gives:

$$p_{\mathrm{R}} = \frac{K_{\mathrm{A}}p_{\mathrm{AR}}}{[\mathrm{A}]} \qquad \text{eq.A10}$$

and

$$\frac{[\mathrm{B}]p_{\mathrm{R}}}{K_{\mathrm{B}}} = p_{\mathrm{BR}} \qquad \text{eq.A11}$$

In fact, we can take this one step further by substituting the expression the expression for p_R from A10 into A11:

$$\frac{[\mathrm{B}]}{K_{\mathrm{B}}} \times \frac{K_{\mathrm{A}}p_{\mathrm{AR}}}{[\mathrm{A}]} = p_{\mathrm{BR}} \qquad \text{eq.A12}$$

Why this is useful we will see in a moment but, first, we have to divide equation A3 by $[R]_T$:

$$\frac{[\mathrm{R}]}{[\mathrm{R}]_{\mathrm{T}}} + \frac{[\mathrm{AR}]}{[\mathrm{R}]_{\mathrm{T}}} + \frac{[\mathrm{BR}]}{[\mathrm{R}]_{\mathrm{T}}} = \frac{[\mathrm{R}]_{\mathrm{T}}}{[\mathrm{R}]_{\mathrm{T}}} \qquad \text{eq.A13}$$

Using the definition of p_R:

$$p_{\mathrm{R}} = \frac{[\mathrm{R}]}{[\mathrm{R}]_{\mathrm{T}}} \qquad \text{eq.A14}$$

We can substitute relations A14, A6 and A7 into equation A13, giving:

$$p_{\mathrm{R}} + p_{\mathrm{AR}} + p_{\mathrm{BR}} = 1 \qquad \text{eq.A15}$$

Now our expression for p_R (A10) and p_{BR} (A12) come in useful because we can substitute them into equation A15 to get:

$$\frac{K_{\mathrm{A}}}{[\mathrm{A}]} \cdot p_{\mathrm{AR}} + p_{\mathrm{AR}} + \frac{[\mathrm{B}]}{K_{\mathrm{B}}} \cdot \frac{K_{\mathrm{A}}}{[\mathrm{A}]} \cdot p_{\mathrm{AR}} = 1 \qquad \text{eq.A16}$$

Multiplying both sides of this equation by [A] gives:

$$K_{\mathrm{A}}\, p_{\mathrm{AR}} \; + \; [\mathrm{A}] p_{\mathrm{AR}} \; + \; \frac{[\mathrm{B}]}{K_{\mathrm{B}}} K_{\mathrm{A}}\, p_{\mathrm{AR}} \; = \; [\mathrm{A}] \qquad \text{eq.A17}$$

Factoring out p_{AR} gives:

$$p_{\mathrm{AR}} (K_{\mathrm{A}} \; + \; [\mathrm{A}] \; + \; \frac{[\mathrm{B}]}{K_{\mathrm{B}}} K_{\mathrm{A}}) \; = \; [\mathrm{A}] \qquad \text{eq.A18}$$

So:

$$p_{\mathrm{AR}} \; = \; \frac{[\mathrm{A}]}{(K_{\mathrm{A}} \; + \; [\mathrm{A}] \; + \; \frac{[\mathrm{B}]}{K_{\mathrm{B}}} K_{\mathrm{A}})} \qquad \text{eq.A19}$$

Finally, factoring out K_A from the first and last terms on the bottom of the left-hand fraction gives the Gaddum equation:

$$p_{\mathrm{AR}} \; = \; \frac{[\mathrm{A}]}{K_{\mathrm{A}} \left(1 + \frac{[\mathrm{B}]}{K_{\mathrm{B}}} \right) + [\mathrm{A}]} \qquad \text{eq.A20}$$